The Role of Bioinformatics in Drug Discovery

A Computational Oncology Perspective: From Sequence Analysis to Targeting KRAS G12D

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Abstract

Bioinformatics has become an indispensable pillar of modern drug discovery, accelerating every stage of the pipeline from target identification to lead optimization. This report examines the algorithmic and infrastructural foundations that underpin computational oncology sequence homology search via BLAST, profile-based multiple sequence alignment via Clustal Omega, curated sequence infrastructure via UniProtKB and Biopython, structural bioinformatics via AlphaFold, and virtual screening / molecular dynamics and applies them to a central case study: KRAS G12D. Long considered undruggable owing to its smooth molecular surface and picomolar GTP affinity, KRAS G12D became a tractable therapeutic target through the bioinformaticsguided discovery of its Switch-II pocket (S-IIP). Two clinical-stage inhibitors illustrate this transformation: MRTX1133, a non-covalent inhibitor selective for the GDP-bound state, and zoldonrasib (RMC-9805), a covalent tri-complex inhibitor targeting the active GTP-bound state. Phase I/II data released in October 2024 showed a 30% objective response rate (ORR) and an 80% disease control rate (DCR) for zoldonrasib in pancreatic ductal adenocarcinoma (PDAC) patients, providing the first clinical proof of concept for pharmacologically targeting active KRAS G12D. The convergence of AlphaFold3, generative-AI ligand design, and network biology is now reshaping the entire paradigm of oncology drug discovery.

Keywords

bioinformatics, drug discovery, KRAS G12D, BLAST, Karlin-Altschul statistics, Clustal Omega, mBed, profile Hidden Markov Models, UniProtKB, Biopython, AlphaFold, CADD, virtual screening, molecular docking, MRTX1133, zoldonrasib, RMC-9805

1. Introduction: Breaking the Dogma of Undruggability

Drug discovery traditionally spans 10 to 15 years and costs upward of USD 2.6 billion per approved molecule. Bioinformatics the interdisciplinary application of computer science, statistics, and biology to the analysis of biological data has fundamentally transformed this process by allowing researchers to computationally explore molecular systems before committing to expensive laboratory experiments. Every step of the process, from the retrieval of a candidate sequence to the ranking of a docked ligand pose, now rests on a stack of algorithms whose statistical assumptions and computational shortcuts directly determine what a bench scientist can and cannot see.

For four decades, KRAS was considered undruggable: it is the most frequently mutated oncogene in human cancer, present in roughly 90% of pancreatic ductal adenocarcinoma (PDAC) cases and about 25% of all human cancers, yet it was deemed pharmacologically inaccessible because of its almost featureless molecular surface and its picomolar affinity for GTP. The emergence of structural bioinformatics and molecular dynamics modeling revealed transient allosteric cavities on this otherwise smooth surface, moving KRAS from a textbook example of an undruggable oncogene to a concrete, structurally characterized therapeutic target a transformation that would have been impossible without computation. This report traces that transformation methodically, starting from the algorithmic layer (Section 3) before reconstructing the specific chain of bioinformatics evidence that led to KRAS G12D-selective inhibitors (Section 4).

2. Bioinformatics Across the Drug Discovery Pipeline

Drug discovery proceeds through well-defined stages, each of which is now substantially augmented and in several cases entirely redefined by computational approaches, as summarized in Table 1.

Table 1. Integration of bioinformatics across the drug discovery pipeline.

Stage	Traditional Approach	Bioinformatics Contribution	Key Tools
Target ID	Phenotypic screens	Genomics, proteomics, network analysis	BLAST, STRING, TCGA
Target Validation	Animal models	MSA, structural conservation, PDB analysis	UniProtKB, Clustal Omega, PDB
Hit Discovery	HTS compound libraries	Virtual screening, pharmacophore modeling	AutoDock Vina, Glide, GNINA
Lead Optimization	Iterative synthesis	Molecular dynamics, free energy calculations	GROMACS, AMBER
ADMET Prediction	In vitro / in vivo assays	ML-based property prediction	SwissADME, DeepPurpose
Clinical Translation	Phase I-III trials	Patient stratification, ctDNA biomarkers	NGS, liquid biopsy

3. Key Bioinformatics Methods in Drug Discovery: Algorithmic Foundations

Beyond naming the tools used at each pipeline stage, it is essential to understand the algorithmic assumptions and mathematical constraints that determine what these tools can reliably detect. The subsections below detail the four algorithmic layers mobilized in this project programmatic sequence infrastructure, homology search, profile-based multiple alignment, and structure/complex prediction before connecting them to virtual screening and molecular dynamics.

3.1 Programmatic Sequence Retrieval and Infrastructure

Every computational oncology workflow ultimately rests on a curated sequence infrastructure. UniProtKB is organized into two complementary sections: Swiss-Prot, a manually reviewed set in which each entry is linked to functional information that has been experimentally validated or expert-curated, and TrEMBL, a much larger unreviewed set annotated automatically by computational pipelines; together they exceeded 227 million sequences as of 2022 [10]. Beyond raw sequence storage, UniProtKB now integrates structural information directly into each entry: more than 85% of entries carry a predicted AlphaFold structure, displayed through an interactive viewer colored by the per-residue confidence metric pLDDT (on a 0-100 scale), alongside experimentally determined PDB structures where available, all unified in the ProtVista annotation viewer [10]. For smallmolecule and reaction-level annotation directly relevant to the drug-discovery focus of this report UniProt standardizes ligand nomenclature through the ChEBI chemical ontology and crossreferences the Rhea database of biochemical reactions; at the time of the 2025 UniProt update, more than twelve thousand curated Rhea reactions were linked to over 28 million UniProtKB protein sequences [11], illustrating the scale at which sequence-tofunction-to-chemistry relationships must now be programmatically managed rather than manually inspected.

Programmatic access to this infrastructure is what makes reproducible, scriptable pipelines possible. Biopython provides the software layer connecting Python-based analysis to this ecosystem: its central data structures are the Seq object, which carries built-in transcription and translation operations, and the SeqRecord object, which attaches an identifier, description, and annotation layer to a sequence. The Bio.Blast module allows a script to query either the NCBI's online BLAST server or a local standalone installation and to parse the resulting XML output programmatically, while the Bio.PDB module provides a parser and toolkit for macromolecular structure files [12]. In this project, this exact stack UniProt REST retrieval plus Biopython SeqIO was used to assemble the five-species KRAS sequence set analyzed in Task 2, underscoring that structural and clinical conclusions drawn later in this report are only as reliable as the retrieval and parsing layer on which they are built.

3.2 Target Homology and the Algorithmic Constraints of BLAST

BLAST does not compute an optimal, exhaustive alignment between two sequences; it approximates the alignments that would maximize a local similarity score by first searching only for short exact matches, or word pairs, of a fixed length w whose local score exceeds a threshold T, and then extending those seed matches outward into full local alignments known as Maximal Segment Pairs (MSPs) [13]. This heuristic is what makes BLAST an order of magnitude faster than earlier full dynamic-programming aligners of comparable sensitivity, but it also means BLAST's behavior is entirely governed by the choice of w and T: lowering T increases the chance of detecting weak, biologically real similarities at the cost of longer runtimes, while raising it accelerates the search but risks missing genuine but divergent homologs a direct trade-off between speed and sensitivity that a user must consciously manage rather than treat as a fixed default [13]. In practice, this trade-off is exposed through named search modes: megaBLAST uses long words (28 bases) and a greedy extension strategy, making it the fastest option and well suited to near-identical sequences, whereas standard BLASTN uses shorter 11-base words and is slower but considerably more sensitive to divergent homologs; PSI-BLAST goes further still by iterating the search and building a position-specific scoring matrix from each round's hits, allowing it to detect distant, otherwise undetectable relationships [14].

Whether an MSP score reflects genuine homology or chance similarity is decided statistically rather than by inspection. Karlin and Altschul's extreme-value theory shows that the probability of observing, purely by chance, an MSP score at or above a given value S follows an exponential distribution governed by two parameters, λ and K, which depend on the scoring scheme and the composition of the database being searched; this is what allows a raw BLAST score to be converted into the E-value reported alongside every hit, giving a principled cutoff between a biologically meaningful alignment and statistical noise [13]. The scoring scheme itself matters directly: protein comparisons in BLAST classically use a PAM-120 substitution matrix, while nucleotide comparisons use a simple match/mismatch scheme, commonly +5 for an identity and 4 for a mismatch [13]. Search scope can also be constrained a priori: the 'Limit by Entrez Query' filter restricts a BLAST search to a chosen organism, a broader taxonomic group, or a curated collection such as RefSeq, which reduces the effective size of the searched database and correspondingly lowers the false-positive rate a filter that becomes essential once a search targets a specific oncogene ortholog rather than the entire non-redundant protein database, nr, or the default nucleotide database, nt [14]. This algorithmic layer is precisely what was mobilized in Task 1 of this project to establish KRAS and p53 homology relationships prior to any structural or clinical interpretation.

3.3 Multiple Sequence Alignment via Profile Hidden Markov Models

Once individual homologs are identified, aligning them simultaneously is what exposes evolutionary constraint at single-residue resolution. Clustal Omega, the tool used for the fivespecies KRAS alignment in Task 2 of this project (Rapport 2 Multiple Sequence Alignment of KRAS) [16], builds its guide tree using the mBed algorithm, whose O(N log N) complexity rather than the $O(N^2)$ complexity of classical pairwise-distance methods is what allows it to scale to virtually unlimited numbers of sequences, reportedly aligning more than 190,000 sequences within hours on a single processor. Once the guide tree is built, the actual alignment step no longer relies on simple substitution-matrix scoring but on profile hidden Markov models aligned to one another through the HHalign algorithm, which is the main source of Clustal Omega's accuracy advantage over older, consistency-based aligners on distantly related sequences [15].

Clustal Omega additionally allows a curated external profile HMM for instance one derived from a Pfam family alignment to be supplied as an External Profile Alignment (EPA) guide for aligning new sequences; across the 94 protein families used to benchmark this feature, mean alignment accuracy rose from 0.627 to 0.653 once Pfam-derived HMMs were applied as external profiles [15]. This detail is directly relevant to a limitation already documented in the companion Task 2 report: the KRAS alignment produced an apparent conservation score of only 0.40 for the G12 position once the two outgroup species (rhesus macaque and zebrafish) were included, an artifact traced not to genuine evolutionary divergence but to N-terminal extensions present in the specific UniProt isoforms retrieved for those two species, which shift the true G12-equivalent position out of register with the mammalian reference. An EPA-guided alignment anchored on a curated Pfam Ras-family HMM, rather than an mBed guide tree built purely from the five retrieved sequences, would be expected to reduce exactly this class of isoform-driven misalignment, since the external profile constrains the placement of poorly conserved or length-heterogeneous termini instead of letting the guide tree infer it from a very small, taxonomically unbalanced sample. This is a methodological refinement worth flagging explicitly for any future re-analysis of the KRAS P-loop across a wider phylogenetic sample.

3.4 Structural Bioinformatics and AlphaFold Integration

Structure-based drug design (SBDD) relies on the three-dimensional structure of the target protein to design molecules that fit its binding pockets with high affinity. AlphaFold2 (DeepMind, 2021), whose creators John Jumper and Demis Hassabis were awarded the 2024 Nobel Prize in Chemistry, predicts protein structures from sequence alone with near-experimental accuracy [7]. AlphaFold3 (2024) extends this capability from single-chain structure prediction to protein-ligand complexes and small molecules, with markedly improved accuracy for protein-molecule interactions a capability that is transformative for allosteric, conformationally dynamic targets such as the KRAS Switch-II pocket, where the relevant druggable state is only transiently accessible and therefore difficult to capture with a single static crystal structure.

The practical integration of these predictions into everyday sequence work is itself a bioinformatics achievement: as noted in Section 3.1, UniProtKB now embeds AlphaFold structures directly into its entries, each colored residue-by-residue according to the pLDDT confidence metric, so that a researcher moving from a BLAST hit to a UniProt entry can immediately assess, region by region, how reliable the predicted fold is before committing computational resources to downstream docking or dynamics a workflow efficiency that did not exist even five years ago [10].

3.5 Virtual Screening, Molecular Docking, and Molecular Dynamics

Virtual screening (VS) computationally evaluates large chemical libraries against a protein target, ranking compounds by predicted binding affinity before any of them are synthesized. Molecular docking algorithms such as AutoDock Vina, Glide, and GNINA predict the binding pose and score of each candidate; GNINA in particular integrates convolutional neural networks (CNNs) into its scoring function, improving accuracy over classical force-field-based scoring alone. Such tools have been applied to identify novel KRAS G12D inhibitor candidates from large compound databases [5].

Molecular dynamics (MD) simulations then model the time-dependent behavior of a protein-ligand system at atomic resolution, revealing binding stability, conformational transitions, and dynamic allostery that are invisible in a single static crystal structure. For KRAS G12D specifically, MD showed that the mutation shifts the Switch-II and alpha3-helix conformational states, increases local residue mobility, and disrupts hydrogen bonds required for GTP hydrolysis. Advanced metadynamics techniques go further, producing free-energy landscapes that reveal hidden druggable pockets and inform mutation-selective inhibitor design rather than generic KRAStargeting strategies that would also engage the wild-type protein.

4. Case Study: Bioinformatics-Guided Targeting of KRAS G12D

4.1 The Undruggable Problem

KRAS was first identified as an oncogene in 1982. For four decades it was considered undruggable: a smooth molecular surface with no obvious hydrophobic binding pocket, combined with a picomolar affinity for GTP/GDP that made competitive nucleotide-site inhibition essentially impossible. The G12D variant compounds this difficulty relative to the more extensively studied G12C variant: Aspartic acid is a weaker nucleophile than Cysteine, which complicates covalent inhibitor design and has historically pushed the field toward non-covalent strategies for this particular substitution.

4.2 Structural Bioinformatics Unlocks the Switch-II Pocket

The breakthrough came from crystallographic and computational studies that identified a cryptic allosteric cavity the Switch-II pocket (S-IIP) which forms specifically in the GDP-bound state and in G12D-specific conformations. Ensemble docking against multiple MD-generated snapshots, rather than a single crystal structure, was what revealed the S-IIP's transient accessibility in the first place; structure-based virtual screening against this dynamic pocket then identified the first S-IIP binders, establishing the structural template on which subsequent inhibitor design was built. It is worth noting that this same principle of cross-referencing multiple sequence and structural sources for validation is exactly what the Task 2 alignment (Section 3.3 above) was designed to test at the sequence level: the P-loop and Switch-II region show full conservation among Human, Mouse, and Rat KRAS, which is the sequence-level counterpart of the structural rigidity that made the S-IIP a viable, mutation-selective target rather than a generic KRAS-wide vulnerability.

4.3 MRTX1133: The First Non-Covalent KRAS G12D-Selective Inhibitor

MRTX1133 [1] was identified through structure-based medicinal chemistry as the first non-covalent, selective KRAS G12D inhibitor, with sub-nanomolar potency (IC50 below 2 nM) and roughly 700-fold selectivity over wild-type KRAS. Its selectivity arises mechanistically because Aspartate-12 forms a salt bridge with MRTX1133's bicyclic amine group an interaction that is structurally absent in the wild-type Glycine context, since Glycine has no side chain capable of forming it.

Strategic Discontinuation.

Despite this scientific validation, development of MRTX1133 was discontinued by Bristol Myers Squibb in the first quarter of 2025 following the Mirati acquisition, a decision attributed to pharmacokinetic instability and portfolio strategy rather than to any scientific failure of the underlying S-IIP-targeting hypothesis.

4.4 Zoldonrasib (RMC-9805): Targeting the Active GTP-Bound State

RMC-9805 represents a paradigm shift by targeting the active, GTP-bound state of KRAS G12D rather than the inactive GDP-bound state addressed by MRTX1133. This choice is mechanistically important because G12D mutants, unlike G12C, retain almost no residual GTP-hydrolysis activity and therefore spend the large majority of their functional cycle in the active state meaning a GDPstate-selective inhibitor such as MRTX1133 can only engage a minority conformational pool of the oncoprotein in cells.

- **Tri-complex mechanism:** RMC-9805 recruits endogenous cyclophilin A (CypA) as a molecular glue, locking KRAS G12D into a conformation that blocks its interaction with the downstream RAF and PI3K effectors.
- **Covalent warhead:** an aziridine warhead reacts covalently and specifically with the Aspartate-12 carboxylate introduced by the mutation, providing mutation-selective targeting that is structurally impossible against wild-type KRAS [2].

Clinical Breakthrough Zoldonrasib Phase I/II (October 2024)

In PDAC patients (n = 40): Objective Response Rate (ORR) 30% | Disease Control Rate (DCR) 80% | ctDNA KRAS G12D reduction greater than 50% in 86% of evaluable patients.

These results constitute the first clinical proof of concept for pharmacologically targeting the active state of KRAS G12D (Papadopoulos et al., 2025) [3].

5. Future Perspectives: PROTACs, Generative AI, and Network Biology

5.1 Resistance Mechanisms and PROTACs

Resistance mutations such as KRAS Y96D reduce S-IIP volume by roughly 33%, preventing inhibitor binding while preserving oncogenic signaling through the RAS pathway. To circumvent this class of resistance, Proteolysis-Targeting Chimeras (PROTACs) such as ASP3082 direct KRAS G12D toward the proteasome for complete protein degradation rather than mere inhibition, which avoids the feedback-loop reactivation that can follow simple active-site or allosteric-pocket blockade [8].

5.2 Generative AI and De Novo Design

Graph neural networks and transformer-based generative models are now applied to the de novo generation of ligands specifically optimized for resistance-associated pockets, substantially shortening the chemistry-biology optimization cycle relative to traditional medicinal chemistry iteration. AlphaFold3's ability to predict protein-ligand co-complexes further enables more accurate ensemble docking for targets, such as KRAS G12D, where several conformational states are simultaneously therapeutically relevant. Combined with ML-based ADMET prediction, this pipeline accelerates the translation from a computational hit to a viable clinical candidate.

5.3 Network Biology and Combination Strategies

Co-mutation of TP53 occurs in approximately 39.4% of KRAS-driven cancers and measurably modulates treatment response, meaning that a purely KRAS-centric computational model is incomplete for a large fraction of real patients. Rational combination strategies targeting both KRAS G12D and downstream signaling nodes MEK, SHP2, or EGFR are consequently being explored both computationally and clinically. Network-based bioinformatics analysis of the KRAS interaction map is what guides the rational selection of synergistic combination partners rather than an empirical trial-and-error search across the full space of possible co-treatments [9].

6. Summary Table of Key Bioinformatics Tools

Table 2 consolidates, for each tool or database mobilized across this project, the specific algorithm underlying it and its concrete role in the KRAS G12D case study, complementing the pipeline-level view given in Table 1.

Tool / Database	Category	Core Algorithm	Primary Use in this Project
BLAST / PSI-BLAST	Sequence homology	Maximal Segment Pair (MSP) heuristic; word-length w / threshold T extension; KarlinAltschul E-value statistics	KRAS/p53 homology search, target identification (Task 1)
UniProtKB	Sequence & annotation database	Swiss-Prot manual curation vs TrEMBL automated annotation; ChEBI/Rhea cross-referencing	Retrieval of the 5-species KRAS sequence set (Task 2)
Biopython	Programmatic infrastructure	Seq / SeqRecord objects; Bio.Blast and Bio.PDB parsers	Scripting of retrieval, parsing, and translation tasks
Clustal Omega	Multiple sequence alignment	mBed guide-tree construction (O(N log N)); HHalign profileHMM alignment; External Profile Alignment (EPA)	429-position KRAS alignment across 5 species (Task 2)
AlphaFold2 / 3	Structure prediction	Deep-learning structure/complex prediction; per-residue confidence (pLDDT, 0-100)	Structural template for the S-IIP pocket
AutoDock Vina / GNINA	Molecular docking	Empirical scoring functions; GNINA adds CNN-based rescoring	Virtual screening for KRAS G12D S-IIP binders

GROMACS / AMBER	Molecular dynamics	Newtonian force-field integration; metadynamics freeenergy sampling	Switch-II / alpha3-helix conformational dynamics
SwissADME / ZINC / ChEMBL	ADMET & compound libraries	Rule-based and ML property prediction; compound source libraries	Lead optimization and hit sourcing

Table 2. Key bioinformatics tools, their core algorithms, and their role in the KRAS G12D case study.

7. Conclusion

Bioinformatics has evolved from a supporting tool into the primary driver of drug discovery. The case of KRAS G12D is the most compelling demonstration of this transformation, and this report has traced it at two levels simultaneously: at the algorithmic level, where MSP-based homology search, profile-HMM multiple alignment, curated sequence infrastructure, and deep-learning structure prediction each impose specific statistical and computational constraints on what can be discovered; and at the applied level, where four decades of failure ended through the systematic use of exactly these methods. Sequence-conservation analysis identified G12 as an ultra-conserved structural constraint; molecular dynamics revealed the dynamic Switch-II pocket; structure-based virtual screening identified the first pocket binders; and iterative computational optimization ultimately produced clinical-grade molecules.

Key Synthesis

Bioinformatics did not merely support KRAS G12D drug discovery it was its enabling foundation. Without the algorithmic machinery detailed in Section 3 Karlin-Altschul statistics to validate homology, mBed/HHalign alignment to expose conservation, and AlphaFold-based structural modeling to resolve the S-IIP no KRAS G12D inhibitor would exist today. The 80% disease control rate achieved by zoldonrasib in PDAC patients (October 2024) is the clinical validation of a computation-first drug discovery paradigm.

As AlphaFold3, generative AI, and ML-based ADMET prediction continue to mature, the gap between bioinformatics prediction and clinical reality will keep narrowing. Mastery of this full algorithmic stack from the E-value statistics behind a single BLAST hit to the tri-complex mechanism behind a Phase I clinical response is the central competence of modern oncology research, and the defining skill of the next generation of computational biologists.

References

1. Wang X, Allen S, Blake JF, Bowcut V, Briere DM, Calinisan A, et al. Identification of MRTX1133, a noncovalent, potent, and selective KRAS(G12D) inhibitor. J Med Chem. 2022;65(4):3123-33.

2. Schulze CJ, Seamon KJ, Zhao Y, Yang YC, Cregg J, Kim D, et al. Chemical remodeling of a cellular chaperone to target the active state of mutant KRAS. Science. 2023;381(6659):794-9.

3. Papadopoulos KP, et al. Preliminary safety and antitumor activity of zoldonrasib (RMC-9805). J Clin Oncol. 2025;43(4_suppl):724.

4. Zeissig MN, Ashwood LM, Kondrashova O, Sutherland KD. Next batter up! Targeting cancers with KRAS-G12D mutations. Trends Cancer. 2023;9(11):955-67.

5. Lu S, Jang H, Gu S, Zhang J, Nussinov R. Drugging specific quaternary structures of KRAS. Trends Cancer. 2021;7(8):734-47.

6. Hallin J, Bowcut V, Calinisan A, et al. Anti-tumor efficacy of a potent and selective noncovalent KRAS G12D inhibitor. Nat Med. 2022;28:2171-82.

7. Jumper J, Evans R, Pritzel A, et al. Highly accurate protein structure prediction with AlphaFold. Nature. 2021;596:583-9.

8. Sobhani N, Pittacolo M, D'Angelo A, Marchegiani G. Recent anti-KRAS G12D therapies: a 'possible impossibility' for pancreatic ductal adenocarcinoma. Cancers. 2025;17(4):704.

9. Mondal K, Posa MK, Shenoy RP, Roychoudhury S. KRAS mutation subtypes and their association with other driver mutations in oncogenic pathways. Cells. 2024;13(14):1221.

10. UniProt Consortium. UniProt: the Universal Protein Knowledgebase in 2023. Nucleic Acids Res. 2023;51(D1):D523-31.

11. UniProt Consortium. UniProt: the Universal Protein Knowledgebase in 2025. Nucleic Acids Res. 2025;53(D1):D609-17.

12. Cock PJA, Antao T, Chang JT, Chapman BA, Cox CJ, Dalke A, et al. Biopython: freely available Python tools for computational molecular biology and bioinformatics. Bioinformatics. 2009;25(11):1422-3.

13. Altschul SF, Gish W, Miller W, Myers EW, Lipman DJ. Basic local alignment search tool. J Mol Biol. 1990;215(3):403-10.

14. McGinnis S, Madden TL. BLAST: at the core of a powerful and diverse set of sequence analysis tools. Nucleic Acids Res. 2004;32(Web Server issue):W20-5.

15. Sievers F, Wilm A, Dineen D, Gibson TJ, Karplus K, Li W, et al. Fast, scalable generation of high-quality protein multiple sequence alignments using Clustal Omega. Mol Syst Biol. 2011;7:539.

16. Hounsou-Kan CAJ. Multiple sequence alignment of KRAS: conservation analysis across 5 species and the G12D oncogenic hotspot [internal report]. Independent Computational Oncology Research Project, Task 2; 2026.